# Speculative Decoding Methods

How different inference methods generate and verify tokens

## 1 Autoregressive Decoding

Single-step generation (no speculation)

**1 Input Prompt / Context**

Where | is | the | capital | of | China?

↓

**Target LLM** (Large Model) — KV Cache

**2 Generate next token (one at a time)**

t: Where is the capital of China? The

↓

t+1: Where is the capital of China? The capital

↓

t+2: Where is the capital of China? The capital of

↓

t+3: Where is the capital of China? The capital of China

**3 Output (all tokens are accepted)**

... of China? The capital of ...

**Summary:** The simplest baseline: reliable and general, but limited by one target-model pass per token.

## 2 Draft-Model Speculative Decoding

Use a small draft model to propose multiple tokens

**1 Input Prompt / Context**

Where | is | the | capital | of | China?

**Draft Model** (Small) — Propose k tokens

↓

The | capital | of | Japan | is

Candidate Proposal (k=5)

**Target LLM** (Large Model) — KV Cache

**2 Verification by target model**

The | capital | of | Japan | is

Accepted — Rejected

↓

Resume from last accepted token

**Target LLM** (Large Model) — KV Cache

**3 Continue generation**

... of China? The capital of ...

**Summary:** Speeds up decoding by proposing multiple tokens with a small draft model, at the cost of extra model and system overhead.

## 3 Prompt Lookup Decoding

Retrieve next tokens from repeated n-grams in prompt

**1 Input Prompt / Context**

Where | is | the | capital | of | China?

**2 Candidate Proposal via n-gram Lookup**

n-gram index (built from prompt)

Matches for prefix "... capital of China?":

- ... China? The capital of China is Beijing.
- ... China? The capital of China remains Beijing.
- ... China? The capital of China, Beijing, is ...

↓

Candidate next tokens (from matches)

The | capital | of | Japan | is

**3 Verification by target model**

**Target LLM** (Large Model) — KV Cache

The | capital | of | Japan | is

Accepted — Rejected

**4 Continue generation from last accepted token**

... of China? The capital of ...

**Summary:** Reuses exact repeated spans from the context without an extra model, but only works well when local repetition is strong.

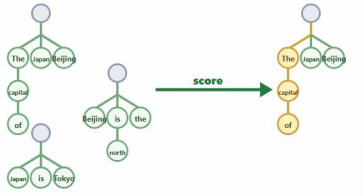
## 4 Suffix Speculative Decoding (Our Focus)

Reuse repeated substrings via suffix matches (no extra model)

**1 Input Prompt / Context**

Where | is | the | capital | of | China?

**2 Candidate Proposal via Suffix Tree**




**3 Verification by target model**

**Target LLM** (Large Model) — KV Cache

The | capital | of | Japan | is

Accepted — Rejected

**4 Continue generation from last accepted token**

... of China? The capital of ...

**Summary:** Unlike draft-model speculation, it needs no extra model, and unlike prompt lookup, it proposes from broader repeated suffix structure rather than only exact local matches.